

Assignment 2

1. (a) Show that the function $f(x) = |x|$, $x \in (-1, 1)$ and zero elsewhere is a possible density function. Find the corresponding distribution function.
- (b) Can the following function be a distribution function?

$$F(x) = \begin{cases} 0, & -\infty < x < 0 \\ \frac{1}{5}, & 0 \leq x < 1 \\ \frac{3}{5}, & 1 \leq x < 3 \\ 1, & 3 \leq x < \infty. \end{cases}$$

If so, find the spectrum and the corresponding probability mass function.

2. A discrete random variable X has the following probability mass function:

$(X = x_i)$	0	1	2	3	4	5	6	7
$P(X = x_i)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

- (a) Find the value of k .
- (b) Evaluate $P(X < 6)$, $P(0 < X < 5)$.
- (c) Determine the distribution function of X .
3. (a) If X is binomial (n, p) variate, find the p.m.f. of $Y = aX + b$, a, b being constants.
- (b) The probability density function of the random variable X is given by

$$f(x) = \begin{cases} 2xe^{-x^2}, & x > 0 \\ 0, & \text{elsewhere.} \end{cases}$$

Find the distribution of X^2 .

4. (a) The probability of hitting a target is 0.001 for each shot. Find the probability of hitting a target with two or more bullets if the number of shot is 2000. (Given that $e^{-2} = 0.1353$) [Ans. $1 - 3e^{-2} = 0.5941$]
- (b) A point X is chosen at random on a line segment AB whose middle point is O. Find the probability that AX, BX and AO form the sides of a triangle. [Ans. $\frac{1}{2}$]
- (c) The I.Q. of students of a class is normally distributed with parameters $m = 100$ and $\sigma = 10$. If the total number of students in the class is 700, then find the number of students who have I.Q. ≥ 115 . Given that $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{1.5} e^{-\frac{t^2}{2}} dt = 0.9332$. [Ans. 47 nearly]
5. (a) If X is normal (m, σ) , then show that $Z = \frac{X-m}{\sigma}$ is a standard normal variate [i.e., Z is $N(0, 1)$ variate].

(b) If X is normal (m, σ) variate, prove that

$$\begin{aligned}P(a < X < b) &= \Phi\left(\frac{b-m}{\sigma}\right) - \Phi\left(\frac{a-m}{\sigma}\right) \\ \text{and } P(|X - m| > a\sigma) &= 2[1 - \Phi(a)]\end{aligned}$$

where $\Phi(x)$ denotes the standard normal distribution function.

6. (a) Find the mean and variance of the Pascal distribution is defined by

$$\begin{aligned}x_i &= i, \quad (i = 0, 1, 2, \dots) \\ \text{and } f_i &= P(X = i) = \frac{1}{1 + \mu} \left(\frac{\mu}{1 + \mu}\right)^i, \quad \mu > 0\end{aligned}$$

(b) Given that the variate X is normal $(0, 1)$, find the variance of e^X .